

AMIT MANJARLY

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SUMMARY

ML Engineer specializing in continual learning, LLM fine-tuning, and real-time inference systems. Deployed end-to-end ML pipelines using PyTorch, Hugging Face, YOLOv8, QLoRA, FastAPI, Docker, and MLflow. Transitioned from 3.5 years of data-backend engineering at Accenture; completed M.S. Data Science at ASU (May 2026).

TECHNICAL SKILLS

ML & AI: Continual Learning, Open-Set Recognition, LLMs (Llama-3.2, Mistral), RAG, Fine-Tuning (QLoRA), CNNs, RNNs
Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, YOLOv8, XGBoost, LightGBM, NumPy, Pandas
Infra & MLOps: Docker, Kubernetes, MLflow, DVC, CUDA, 4-bit Quantization, ChromaDB, FastAPI, Flask, Streamlit
Languages & DBs: Python (Advanced), SQL, C++; Hive, Impala, Oracle, SQLite

AI & MACHINE LEARNING PROJECTS

Open-World Continual Learning for Autonomous Agents [\[github\]](#)

Jan 2026 – April 2026

CNN · Continual Learning · Open-Set Recognition · Domain Adaptation | PyTorch · YOLOv8 · EWC · FastAPI · MLflow

- Fine-tuned YOLOv8m on Waymo Open Dataset; applied EWC ($\lambda=0.4$) on nuScenes to prevent catastrophic forgetting.
- Entropy-based OOD thresholding flags unknown objects at inference time, eliminating silent misclassification.
- Deployed FastAPI + Streamlit via docker-compose; MLflow and DVC track experiments, checkpoints, and data versions.
- Achieved **<120ms** CPU inference latency; modular design supports plug-in detection heads for new object classes.

Nexus - Continual Knowledge Agent (Mistral AI Hackathon 2026) [\[github\]](#)

Mar 2026

RAG · Vector Embeddings · LLM Inference · Two-Stage Retrieval | Flask · ChromaDB · Mistral · Tavily · JWT · SQLite

- Full-stack RAG agent ingesting PDFs, URLs, and text into per-user ChromaDB stores via mistral-embed embeddings.
- Two-stage retrieval (vector DB to Mistral validation to Tavily fallback) yields relevant context on **>90%** of queries.
- Secured with PBKDF2-SHA256 (260k iterations) plus JWT auth; per-user SQLite isolation for all stored data.
- Shipped Flask REST API plus single-page HTML frontend with live usage stats, memory browser, and account management.

LLM Fine-Tuning for Mathematical Reasoning (STaR) [\[github\]](#)

Sep 2025 – Oct 2025

LLM Fine-Tuning · Chain-of-Thought · Self-Taught Reasoning · PEFT | Llama-3.2-3B · QLoRA · bitsandbytes · Hugging Face

- Fine-tuned Llama-3.2-3B on GSM8K via the STaR loop; outperformed Zero-Shot CoT (**47.56%**) and Vanilla SFT (**44.00%**).
- Vanilla SFT underperformed zero-shot CoT due to overfitting; STaR was adopted to build a cleaner rationale dataset.
- STaR bootstrapping verifies rationales and re-prompts failures with ground truth to improve training data quality.
- 4-bit QLoRA + bfloat16 cuts GPU memory by **~75%**, enabling single-GPU fine-tuning at **approx 3x** speed vs. full precision.

End-to-End Hybrid Recommendation Engine [\[github\]](#)

May 2025 – Jun 2025

Collaborative Filtering · SVD · Content-Based Filtering · Cold-Start | Scikit-learn · TF-IDF · Pandas · MovieLens 100k

- SVD collaborative filtering on MovieLens 100k: Precision@10 **0.26** vs. 0.15 (popularity), a **73%** relative gain.
- Automated latent-factor tuning via Precision@10 validation curve; TF-IDF cosine similarity covers cold-start users.
- Hybrid blending layer prioritizes SVD personalization over content-based results, scaling with interaction density.

PROFESSIONAL EXPERIENCE

Accenture PLC

Mar 2021 – Jul 2024

Software Engineer — Data & Backend — Hyderabad, India

- Built Python/SQL automated testing framework for an international banking platform, cutting QA cycle time by **40%**.
- Optimized Hive/Impala SQL pipelines processing millions of daily transactions, reducing end-to-end latency by **33%**.
- Refactored core ETL scripts, cutting production processing time by **15%**; managed data across Hive, Impala, Oracle.
- Automated data quality checks with Python and SQL, improving reliability in model training data pipelines.
- Increased data processing efficiency by **25%** by analyzing complex workflows and implementing targeted fixes.
- Built Tableau dashboards for pipeline health monitoring; coordinated cross-functional delivery via JIRA and Kanban.

EDUCATION

M.S. in Data Science, Analytics and Engineering

Aug 2024 – May 2026

Arizona State University — Tempe, AZ Coursework: Machine Learning, Deep Learning, NLP, Statistical Inference

B.E. in Computer Engineering

Aug 2015 – Jun 2019

Gujarat Technological University — Ahmedabad, India Coursework: DSA, OS, DBMS, Computer Networks, OOP